

Introduction

The aim of this project was to develop a four-page informational website exploring the history, present, and future of human space exploration. The focus was on creating a clear, accessible, and visually consistent user experience while applying modern web development practices such as semantic HTML, modular CSS, responsive design, and accessibility considerations. Throughout the development process, I iteratively refined the layout, styling, and structure of each page, addressing challenges as they emerged and improving my understanding of CSS specificity, layout techniques, and maintainable code organisation.

Website Purpose, Target Audience and Design Choices

The purpose of the website is to inform users about the history of space travel, as well as present and future human space activities, through a clear, accessible, and visually engaging interface. The site is aimed at a general audience with an interest in space exploration, including students, casual learners, and individuals seeking an overview of the major milestones and upcoming missions.

The design prioritises readability, accessibility, and consistency across pages. The global navigation bar and footer provide a consistent structure, while each page uses a `<main>` container to ensure predictable layout behaviour. The colour scheme uses dark blue for headings and accents, reflecting the theme of space and providing strong contrast against a white background. Typography is kept simple and readable using Arial, chosen for clarity and accessibility. Finally, accessibility features such as alt text for images and ARIA labels are used throughout to improve accessibility for people using screen readers.

The web pages follow a minimalistic yet eye-catching design. The history, future, and contact pages use centred content for a clean, streamlined design, while the home page uses a simple two-column layout to introduce the site's purpose alongside a relevant image.

Key Technologies and Design Decisions

HTML5 semantic elements (`<nav>`, `<main>`, `<footer>`, `<div>`, etc.) were used to create a clear document structure and improve accessibility. The main layout of each page is wrapped in the `<main>` tag, while `<nav>` and `<footer>` are used globally to provide consistent navigation and layout. CSS was used extensively for layout, including flexbox for the homepage's two-column layout and CSS grid for the history page's photo gallery. The future page uses a more complex layout based on absolute positioning and pseudo-elements to create a vertical timeline with alternating event blocks.

CSS transitions were used for interactive elements such as hover effects on images. The circular images on the timeline on the future page were created by setting the width and height to equal, fixed dimensions (300px) combined with `'object-fit:cover'` and `'border-radius: 50%'` to create the rounded corners. `'overflow:hidden'` was used for overflow handling, for example to prevent zoomed images from escaping their containers. Responsive design considerations included percentage-based widths, auto-scaling images, and a media query to adapt the timeline layout for smaller screens.

As well as an individual stylesheet for each page, a global stylesheet was used to centralise shared styles for the navigation bar and footer, ensuring consistency and removing the need for duplication. Page-specific stylesheets were used for individual layouts to keep the code maintainable and organised.

Challenges Faced and How They Were Resolved

One of the main challenges was creating layouts that remained consistent and responsive across different pages. Early attempts at the homepage layout used a `<table>`, which caused spacing issues and did not adapt well to different screen sizes. This was resolved by switching to a CSS-based two-column layout using

flexbox, which provided much greater control. This challenge taught me the importance of separating structure and styling. I decided to use one HTML file and one CSS file for each page, to keep structure and styling for each page separate, as well as keeping the styling separate between pages.

The timeline on the future page presented a more complex challenge. Positioning the central line, alternating containers, and circular markers required careful use of absolute positioning and pseudo-elements. Ensuring that the layout remained readable on smaller screens required a dedicated media query to collapse the alternative structure into a single-column layout.

At first, I didn't have many issues with the timeline, as I had used a template from W3Schools which proved fairly easy to adapt to my needs. However, the introduction of the navigation bar created an unexpected conflict. I noticed that the navigation bar on the future page took up less space vertically compared to the other pages. I decided to go through each CSS file, looking for any scoping issues. For example, I changed the scope of a global 'img' selector on the history page to '.content img'. The '.container' class was also used both in the future page timeline and the history page gallery, therefore, I renamed the gallery container to 'history-grid' to prevent the styles from interfering with each other. Finally, I realised that I had given the individual stylesheets higher priority than the global stylesheet, therefore, I reordered the <link> elements so that global.css loaded first on each page. However, none of these attempts resolved the issue with the navigation bar.

After many failed attempts to resolve the issue, I realised that I had initially used an asterisk to select every element on the future page, and apply 'box-sizing: border-box;' to them. However, this unintentionally also applied to the navigation bar, causing it to take up less space vertically. Although I had kept the styling for the navigation bar separate to the styling for each individual page, using a 'global.css' file, this wasn't enough to prevent conflicts between the stylesheets. At first, I tried to resolve this issue by simply removing the part of code applying 'box-sizing: border-box' to the elements, however, this significantly affected the layout of the timeline, causing the vertical line to no longer be centred and to go through some of the boxes. I realised that the issue could not be resolved by just removing this code. I had to keep it in but change the scope. I finally realised that I had to apply 'box-sizing: border-box' to just '.timeline', '.timeline *', '.timeline *::before', '.timeline *::after'. That centred the vertical line as intended, however, the circles on the timeline were still slightly off-centre.

These challenges collectively improved my understanding of CSS specificity, layout techniques, responsive design, and the importance of maintaining clear, modular stylesheets.

Conclusion

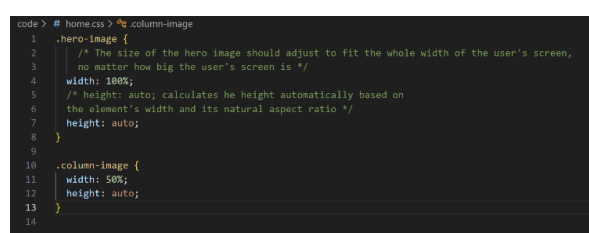
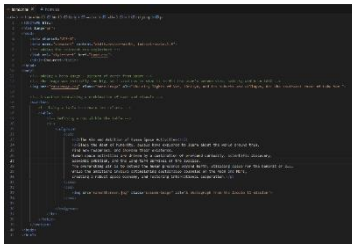
Developing this website allowed me to apply a wide range of web development techniques while gaining practical experience in structuring, styling, and debugging multi-page layouts. Through iterative refinement, I learned how to manage CSS specificity, maintain consistent design across pages, and resolve conflicts between global and page-specific styles. The project strengthened my understanding of responsive design, semantic HTML, and accessibility best practices. Overall, the development process improved both my technical skills and my ability to design clear, maintainable, and user-friendly web interfaces.

Development log:

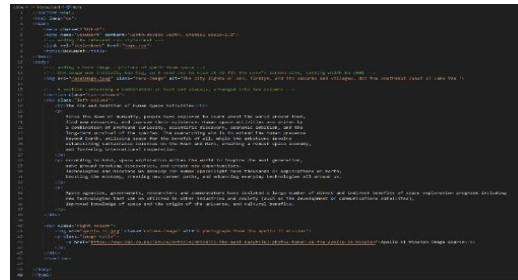
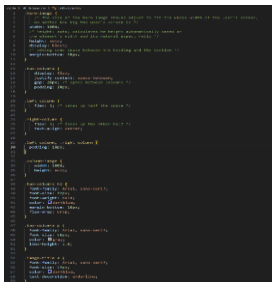
27 January 2026 - 11:20 to 13:20 (2 hours) - Homepage setup and early layout attempts

I created the homepage structure and added a responsive hero image using width: 100%.

My first attempt at creating a two-column layout used a <table>, but this caused spacing and responsiveness issues. I replaced it with a CSS-based two-column layout using a container and flexbox, which gave better control and reinforced the importance of separating structure and styling.



Initial attempt using a table



Using CSS to split the content into two columns

27th January 2026 - 22:30 to 23:50 (1hr20mins) - Began working on history.html



I began building history.html by creating <div> elements with the class set to “event-card, each containing a heading, paragraph, and image. This established a consistent structure for later styling.

5th February 2026 - 11:00 to 11:45 (45 mins) – Began working on future.html



I started future.html by creating a series of event sections, each containing a heading and image. These were prepared for later conversion into a timeline layout.

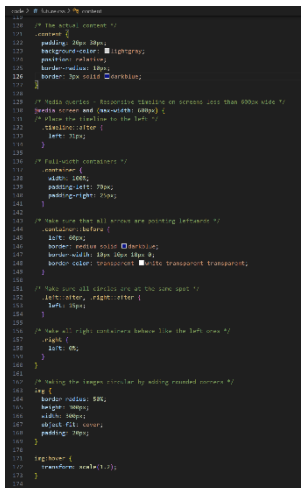
6th February 2026 – 9:40 to 12:10 (2h30mins) – Continued work on future.html and future.css

Using a W3Schools timeline template as reference, I built the full timeline structure.

I created .content boxes nested inside alternating left/right .container elements, all wrapped in a .timeline container. I added a vertical line, circular markers, and directional arrows.

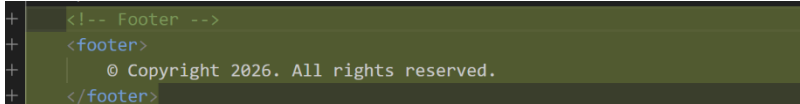
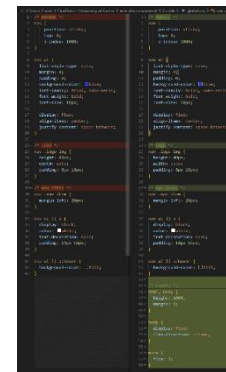
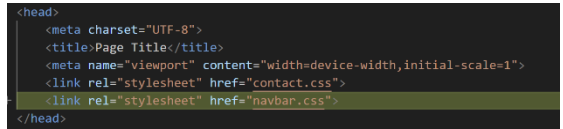
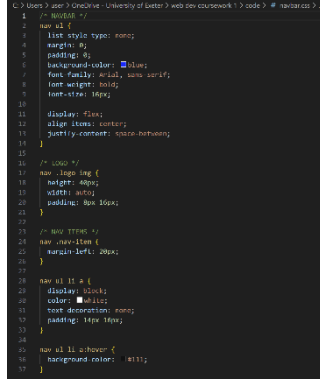
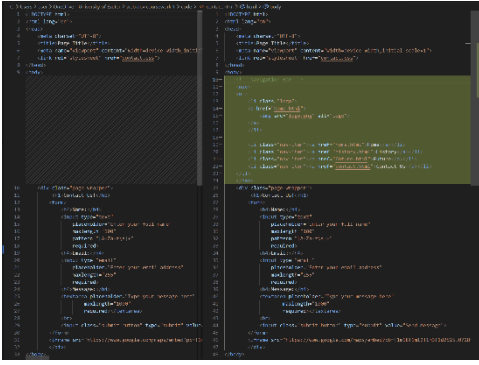
I created circular images by setting an equal width and height, using “object-fit: cover” to make the images square at first, then using “border-radius:50%” to round the corners, creating circles.

Hover effects were added using transform:scale(1.2).



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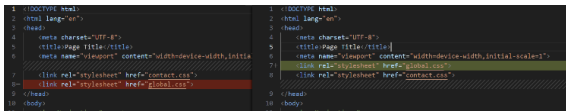
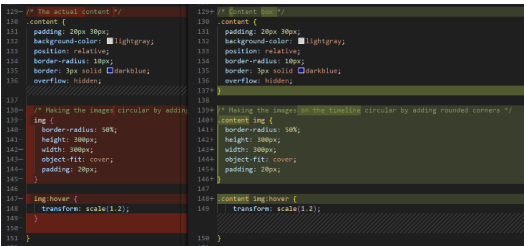
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10th February 2026 – 22:00-00:00 (2 hours) – Adding Navigation Bar and Footer

I added the navigation bar HTML code to each HTML page, then created a navbar.css file to style it.

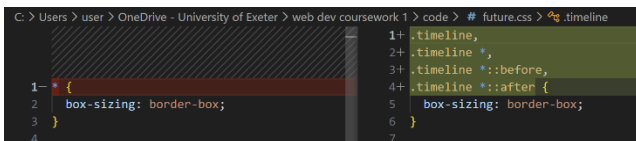
I later realized it would make more sense to create a `global.css` file with both the CSS for the navigation bar and the footer, so I renamed the `navbar.css` file and added the footer styling. I applied the global stylesheet to each HTML file, however, mistakenly gave it lower priority than the other CSS stylesheets.

12th February 2026 – 19:00 to 20:00 (1 hour) – Debugging navbar issues



I noticed that the navigation bar on the future page took up less space vertically compared to the other pages. I tried fixing this issue by changing scoping (e.g. changing 'img' in future.css to '.content img') and giving global.css higher priority than the other (page-specific) stylesheets, however, this still didn't resolve the issue.

13th February 2026 – 14:00-15:30 (1hr30mins) – Fixing navbar issue



I reviewed the navigation bar inconsistency on the future page, and identified that the issue was caused by using a universal selector (*) in future.css to apply box-sizing: border-box to every element on the page. This

unintentionally affected the navbar and changed its height. Removing the rule broke the timeline layout, so I instead fixed the issue by scoping the box-sizing rule specifically to the timeline elements. This preserved the timeline layout while preventing the navbar from being affected.

References:

For researching HTML/CSS:

https://www.w3schools.com/html/html_images.asp

https://www.w3schools.com/howto/howto_css_hero_image.asp

<https://www.educative.io/answers/how-to-create-columns-in-html>

https://www.w3schools.com/howto/howto_css_two_columns.asp

https://www.w3schools.com/html/html_classes.asp

https://www.w3schools.com/howto/howto_css_timeline.asp

https://www.w3schools.com/howto/howto_css_rounded_images.asp
https://www.w3schools.com/html/html_forms.asp
https://www.w3schools.com/css/css3_buttons.asp
<https://stackoverflow.com/questions/69077266/how-to-add-2-rows-and-3-columns-images>
https://www.w3schools.com/Css/css_navbar_horizontal.asp
https://www.w3schools.com/tags/tag_footer.asp

For researching space:

https://en.wikipedia.org/wiki/Sputnik_1
<https://en.wikipedia.org/wiki/Laika>
https://en.wikipedia.org/wiki/Yuri_Gagarin
<https://www.nasa.gov/image-article/april-1961-first-human-entered-space/>
<https://www.planetary.org/space-missions/vostok-1>
<https://www.britannica.com/topic/Apollo-11>
<https://www.nasa.gov/history/apollo-11-mission-overview/>
<https://airandspace.si.edu/explore/stories/apollo-11-moon-landing>
<https://www.britannica.com/topic/International-Space-Station>
<https://issnationallab.org/about/iss-national-lab-overview/iss-history-timeline/>
https://en.wikipedia.org/wiki/Alexei_Leonov
https://en.wikipedia.org/wiki/Valentina_Tereshkova
<https://washingtondc.jhu.edu/news/space-outlook-2026/>
https://en.wikipedia.org/wiki/Artemis_II
<https://www.nasa.gov/mission/artemis-iii/>
<https://www.nasa.gov/mission/nasas-spacex-crew-12/>

Image sources:

Website Logo:

<https://www.istockphoto.com/vector/planet-doodle-gm486905364-73801987>

I used this image for the logo in the navigation bar, removing the background using Canva.

Home Page:

Hero image: [https://www.nasa.gov/image-article/night-lights-of-bangkok-capital-of-thailand/#:~:text=iss064e037842%20\(Feb.%2026%2C%202021\)%20%E2%80%94%20The%20night,miles%20](https://www.nasa.gov/image-article/night-lights-of-bangkok-capital-of-thailand/#:~:text=iss064e037842%20(Feb.%2026%2C%202021)%20%E2%80%94%20The%20night,miles%20)

[above%20the%20southern%20coast%20of%20Vietnam.](#)

Apollo 11 image: <https://www.bbc.co.uk/future/article/20190715-the-most-beautiful-photos-taken-on-the-apollo-11-mission>

History page:

Event cards:

Sputnik 1: <https://spacenews.com/americas-next-sputnik-moment-is-already-here/>

Laika: <https://www.mirror.co.uk/news/world-news/first-dog-sent-to-space-35002909>

My grandma meeting Yuri Gagarin:

<https://www.vtedy.sk/galeria/rok-1961:-sovietiski-vedci-vyslali-do-kozmu-prveho-cloveka/9?currentPhoto=5>

Alan Shepard: <https://www.nasa.gov/image-article/celebrating-astronaut-alan-shepards-100th-birthday/>

Alexei Leonov: <https://www.theguardian.com/science/2019/oct/11/alexei-leonov-space-walk-human-death-cosmonaut-russia>

Valentina Tereshkova: <https://www.space.com/21571-valentina-tereshkova.html>

Apollo 11: <https://www.astrotourism.com/news/moon-landing-50-years-after-apollo-11/>

International Space Station: <https://www.space.com/international-space-station-future-end-congress-ask-nasa>

Photo gallery:

Sputnik 1: <https://astrobiology-nz-ecly.fixed-staging.co.uk/sputnik-1-and-the-space-age/>

Sputnik 2: https://en.wikipedia.org/wiki/Sputnik_2

Sputnik 2 postage stamp: https://en.wikipedia.org/wiki/Sputnik_2

Laika: <https://www.space.com/laika-space-dog>

Yuri Gagarin: <https://www.space.com/16159-first-man-in-space.html>

Alan Shepard: <https://www.nasa.gov/image-article/astronaut-alan-shepard/>

Alexei Leonov: <https://www.theguardian.com/science/gallery/2015/may/09/first-space-walk-alexei-leonov-in-pictures>

Valentina Tereshkova: <https://nmspacemuseum.org/inductee/valentina-v-tereshkova/>

Apollo 11 lift off: <https://blog.sciencemuseum.org.uk/buzz-aldrin-in-conversation-with-brian-cox/>

Apollo 11 moon landing: <https://science.nasa.gov/resource/apollo-11-mission-image-astronaut-edwin-aldrin/>

Inside the International Space Station:

<https://www.theguardian.com/artanddesign/gallery/2020/nov/10/gimme-some-space-inside-the-international-space-station-in-pictures>

International Space Station: <https://kids.britannica.com/students/assembly/view/185622>

Future Page:

SpaceX Crew-12: <https://www.nasa.gov/mission/nasas-spacex-crew-12/>

Artemis II: https://en.wikipedia.org/wiki/Artemis_II

Shenzhou 23: https://en.wikipedia.org/wiki/Shenzhou_23spacecraft

Artemis III: <https://www.nasa.gov/missions/artemis/artemis-iii/>

AI Declaration:

No generative AI tools were used in the development of this coursework.

